Learning Dynamic Stream Weights For Coupled-HMM-Based Audio-Visual Speech Recognition

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***Abstract*—With the increasing use of multimedia data in communication technologies, the idea of employing visual information in automatic speech recognition (ASR) has recently gathered momentum. In conjunction with the acoustical information, the visual data enhances the recognition performance and improves the robustness of ASR systems in noisy and reverberant environments. In audio-visual systems, dynamic weighting of audio and video streams according to their instantaneous confidence is essential for reliably and systematically achieving high performance. In this paper, we present a complete framework that allows blind estimation of dynamic stream weights for audio-visual speech recognition based on coupled hidden Markov models (CHMMs). As a stream weight estimator, we consider using multilayer perceptrons and logistic functions to map multidimensional reliability measure features to audiovisual stream weights. Training the parameters of the stream weight estimator requires numerous input-output tuples of reliability measure features and their corresponding stream weights. We estimate these stream weights based on oracle knowledge using an expectation maximization algorithm. We define 31-dimensional feature vectors that combine model-based and signal-based reliability measures as inputs to the stream weight estimator. During decoding, the trained stream weight estimator is used to blindly estimate stream weights. The entire framework is evaluated using the Grid audio-visual corpus and compared to state-of-the-art stream weight estimation strategies. The proposed framework significantly enhances the performance of the audio-visual ASR system in all examined test conditions.**

***Index Terms*—Audio-visual speech recognition, coupled hidden Markov model, logistic regression, multilayer perceptron, reliability measure, stream weight.**

I. Introduction

ALTHOUGH statistical models, like the hidden Markov models (HMMs), have shown great success in modeling and recognizing speech signals under lab conditions, noise and reverberation preclude wider use of ASR systems. This is due to the massive corruption of speech signals in real-world environments, which leads to a rapid degradation in the ASR performance under adverse acoustical conditions [1]. A range of front-end and back-end methods [2], [3] have been proposed in order to improve the ASR performance in the presence of noise. One of these methods that has recently attracted research interest is using visual features encoding the appearance and shape of the speaker's mouth in conjunction with the conventional acoustical features. The motivation of this approach is that the visual features are independent of the acoustical environment while relevant to the speech production process.

In order to model the speech production process using both the acoustical and visual information, many models have been proposed. These models differ regarding the point where the fusion of the audio and video streams takes place. For example, in direct integration (DI) models, the fusion is applied on the feature level by simply concatenating the audio and visual features [4], or by combining the features in a more complex manner using techniques like dominant or motor recording [5], [6]. Alternatively, separate integration (SI) models [6], [7] integrate the audio and video modality at the classifier output level. The fusion level in SI models varies according to the definition of the classifier output, e.g., phrase, word, or phoneme level.

Between these two extreme integration models, there are the state-based decision fusion (SBDF) models, e.g., the fully synchronized multistream HMM (MSHMM) [8]–[12]. In this paper we use an SBDF model called the coupled HMM [9]–[11], [13]. CHMMs have the advantage of allowing asynchrony on the state level while preserving the natural dependency between the audio and video modality by forcing synchronization at certain speech units (here, at word boundaries).

In contrast to the DI models, both SI and SBDF models have the capability to deploy so-called stream weights. Depending on the reliability of each modality, which varies according to its information content and the time-varying environmental influences, the stream weights control the contribution of each modality to the final decision. For example, if the acoustical environment is very noisy, the acoustical features will be severely corrupted and they should contribute less to the total score than the visual features, as the latter are almost independent of the acoustical environment. On the other hand, in clean acoustical environments, the acoustical features should contribute more to the total score than the visual features, since they contain more phonetic information.

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While in some prior works, the stream weight for the whole dataset has been set to a fixed value, which was found using grid search, e.g., [11], [14] or using other tuning algorithms, e.g., [15], some authors have assumed that the stream weight is a model parameter and have estimated it using generative [16] or discriminative [17], [18] criteria. In real scenarios, however, the reliability of the audio and video modality can vary quickly, even on the frame level, and such fixed or model-dependent estimation might lead to worse results than using Bayes fusion, i.e., equal weights. Therefore, in this paper we propose a framework that blindly estimates optimal stream weights for each time frame.

For this purpose, we consider training frame-dependent stream weight estimators in a supervised manner as a regression problem. Hence, it can be split into three sub-problems:

1) Estimating the ideal (*oracle*) stream weights, i.e., the training targets
2) Choosing the input features used as arguments for the regression function
3) Choosing a function suitable to optimally map the input features to the dynamic stream weights

The proposed framework involves a solution for each of these three sub-problems: As a solution for the first sub-problem, we present an expectation maximization (EM) algorithm that estimates oracle dynamic (frame-dependent) stream weights for CHMM-based AVASR. The estimated stream weights using the proposed EM algorithm are termed *oracle* because prior knowledge of the correct word sequence enters into their computation. Hence, they are only usable during training but not during test time. The large number of oracle stream weights provided by this algorithm makes it possible to use multidimensional reliability measure feature vectors, which combine different signal-based and model-based reliability measures, as arguments for the regression function. The proposed multidimensional reliability measure feature vectors contain the so-called *acoustical feature uncertainties*, soft and hard voice activity detection cues, and all reliability measures used in [19], e.g., the entropy and the signal-to-noise ratio. Finally, addressing the third problem, we consider two different mapping functions, multilayer perceptrons or logistic functions, to blindly estimate the stream weights using the proposed reliability measure feature vectors.

The remaining paper is structured as follows: After discussing the relation of the proposed framework to prior works in Section II, we give a brief overview of the CHMM in Section III. Next, in Section IV, an EM algorithm for estimating the oracle dynamic stream weights is introduced. Further, a greedy optimization strategy that estimates a reasonable initialization for the oracle stream weight vectors is shown and a summary of the entire algorithm is given. In Section V, we introduce the definitions of some reliability measures that have previously been used in works like [19], and of other reliability measures that are newly proposed. In Section VI, we discuss some investigations done for choosing the mapping function. The entire framework is summarized in Section VII. Experiments and results are described in Section VIII, which is structured as follows: After introducing the experimental setup in Section VIII-A, the EM-based oracle dynamic stream weights are evaluated in Section VIII-B. In Section VIII-C, we introduce the baseline results that are obtained using [19]. Having established this reference, we compare the results obtained using the proposed framework to the best results in the baseline in Section VIII-D. Finally, we conclude the paper and give an outlook on further work in Section IX.

II. Relation to Prior Works

The three sub-problems introduced in Section I have been solved as follows in [19]: The oracle weights have been set to global fixed values, which have been obtained empirically for each noise type and level, i.e., for each available dataset, by a grid search. The global stream weights are assumed to be mapped from one-dimensional reliability measures, e.g. the signal-to-noise ratio (SNR), using a second order exponential function. In order to train the parameters of this mapping function, the global fixed stream weights have been used as the oracle target outputs of the mapping function and the average of all reliability measures in the corresponding dataset has been considered as the corresponding input. During recognition, the mapping function is employed in each frame, and can thus take temporal variations of the stream reliability into consideration even on the segment-level.

One problem of this approach is that the number of input-output tuples used for estimating the mapping function parameters equals the number of datasets, which is very small. Moreover, the parameters of the mapping function have been learned from fixed stream weights, although the mapping function is intended to be deployed on a frame-by-frame basis. In our approach, we have avoided these limitations by estimating oracle frame-dependent stream weights using the expectation maximization algorithm introduced in Section IV. Another issue of the approach introduced in [19] is that only one-dimensional reliability measures of the audio stream have been considered as input features, in effect considering the video data to be of constant reliability and utility. For our data, however, we have observed strong changes in the reliability of video signals across different speakers, and we have not found a single reliability measure for completely assessing the quality of the audio signal. We are therefore considering 31-dimensional reliability measure feature vectors that include signal-based and model-based reliability measures extracted from both the audio and the video stream.

Unlike [19], the authors of [9], [20] have considered video and audio reliability measures as input features of a multidimensional sigmoid function. The parameters of this function have been estimated so that a proposed discriminative function is maximized. This approach needs frame-level labels for audio-visual (AV) observations. Those labels have been found by applying the forced alignment algorithm to the clean speech features alone. The obtained frame-state alignments have been used as the required labels for the audio-visual states. The authors of [21] claimed that this type of alignment may be inconsistent for CHMMs and re-estimation of frame-state alignment is mandatory. The expectation maximization algorithm deployed in our framework has enabled us to avoid this problem by estimating new two-dimensional state occupation probabilities in every EM iteration.

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In [22], two approaches have been proposed to train the stream weight estimator. In the first one, the reliability measures have been firstly clustered into N clusters. The averages of these clusters have then been used with the oracle stream weights found via grid search to train the parameters of a sigmoid function. This first approach is very similar to [19] and therefore has its limitations. The second approach models the conditional probability of the reliability measures—given the oracle stream weight again determined by a grid search—as a full covariance Gaussian mixture model (FCGMM). The stream weights are then estimated either as the conditional mean of the stream weight given the reliability measures or as the argument that maximizes the posterior probability of the stream weight given the reliability measures. In addition to the problem of using global fixed weights, this approach may be intractable by considering the stream weight as a continuous quantity, as stated by the authors [22]. We have avoided this problem in the proposed framework by using the MLPs or logistic functions instead of the FCGMM in order to capture the variability and the correlation of the reliability measure features given a continuous (non-quantized) stream weight.

Compared to our two recent methods in [23], [24], this paper contains detailed and rigorous mathematical derivations of the training approaches, see Appendix A. Moreover, we compare the results obtained from our greedy initialization approach with the results of the proposed EM algorithm and provide the development set results of all experiments. This comparison shows the need of further smoothing the stream weights obtained from the initialization algorithm by using the proposed EM algorithm.

Since the form of the mapping function is generally unknown [22], the regression functions have been heuristically chosen in the related prior works, including [19], which is adopted here as our baseline. In Sections VI and VIII-C, we provide new analyses and experimental results in order to investigate which type of mapping function is optimal for blindly estimating stream weights.

III. Coupled Hidden Markov Models

The CHMM is an audio-visual fusion model that takes into account the asynchronicities between the visible articulator movements and the acoustic voice production, while enforcing their synchronization at word boundaries. A CHMM consists of a 2-dimensional matrix of composite states q. The CHMM composite states are composed of the marginal audio state q_A and the marginal video state q_V, cf. Fig. 1. Each composite state q has an emission score

$$b_{q,\lambda}(O) = \mathrm{p}(O_A|q_A)^{\lambda}\mathrm{p}(O_V|q_V)^{1-\lambda}, \tag{1}$$

where $\mathrm{p}(O_A|q_A)$ and $\mathrm{p}(O_V|q_V)$ are the audio- and video-only state-conditional observation likelihoods, respectively, and $0 \leq \lambda \leq 1$ is the stream weight, which reflects the relative reliability of the acoustical observation O_A compared to the visual observation O_V. Note that the audio-visual observation vector is defined as $O = \{O_A, O_V\}$. The transition probability between two composite states $i = (i_A, i_V)$ and $j = (j_A, j_V)$ in a CHMM can be written as

$$a_{i,j} = \mathrm{p}\left(q_{t+1} = j | q_t = i\right) = \prod_{s\in\{A,V\}} a^s_{i_s,j_s}, \tag{2}$$

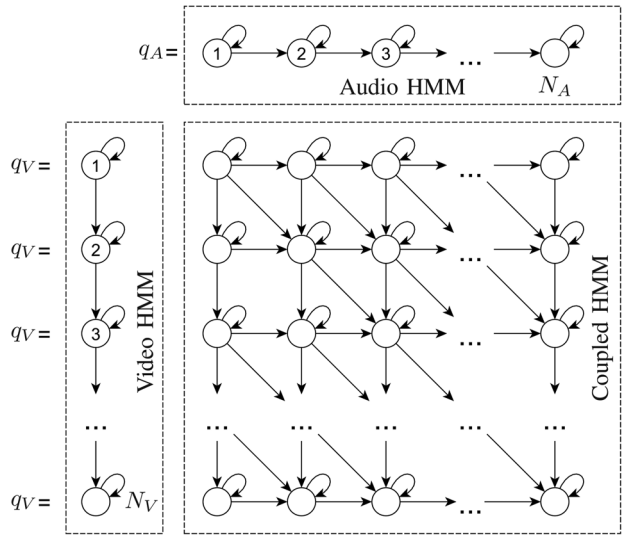


Fig. 1. Coupled HMM with $N_A \times N_V$ composite states.

where $a^s_{i_s,j_s}$ is the transition probability of the corresponding states of the single-stream HMMs.

Since the parameters of the marginal single stream HMMs are trained separately, the number of CHMM parameters increases linearly, although the number of the CHMM states increases with $N_A \times N_V$ (cf. Fig. 1). Therefore, the log-likelihood computation overhead also increases only linearly. This can be thought of as parameter tying [9].

Another advantage of using the CHMM as a fusion model is that the number of states of the marginal models can be chosen independently to match different sampling rates and other recording conditions of different modalities. Moreover, each marginal model can be replaced at any time with a better model without retraining all model parameters.

IV. Oracle Dynamic Stream Weights

A. Expectation Maximization Algorithm

In the following, by a bold symbol quantity we denote a sequence of this quantity. For example, by $\boldsymbol{\lambda}$ and $\boldsymbol{O}$, we denote a sequence of oracle dynamic stream weights and a sequence of audio-visual observations, respectively.

We estimate the oracle dynamic stream weights $\boldsymbol{\lambda} = \{\lambda_t\}_{t=1}^T$ given an audio-visual observation sequence $\boldsymbol{O} = \{O_t\}_{t=1}^T$ and the corresponding true word sequence $\boldsymbol{w}$ as

$$\hat{\boldsymbol{\lambda}} = \arg\max_{\boldsymbol{\lambda}} \left\{\mathrm{p}\left(\boldsymbol{O}, \boldsymbol{\lambda}|\boldsymbol{w}\right)\right\}. \tag{3}$$

As shown in [23], the joint probability $\mathrm{p}(\boldsymbol{O}, \boldsymbol{\lambda}|\boldsymbol{w})$ in (3) can be computed by marginalizing the joint probability $\mathrm{p}(\boldsymbol{O}, \boldsymbol{q}, \boldsymbol{\lambda}|\boldsymbol{w})$ over all possible 2-dimensional state sequences $\mathcal{Q}$. This means that

$$\mathrm{p}\left(\boldsymbol{O}, \boldsymbol{\lambda}|\boldsymbol{w}\right) = \sum_{\boldsymbol{q}\in\mathcal{Q}} f_{\boldsymbol{q}}(\boldsymbol{\lambda}), \tag{4}$$

where

$$f_{\boldsymbol{q}}(\boldsymbol{\lambda}) = \mathrm{p}\left(\boldsymbol{O}, \boldsymbol{q}, \boldsymbol{\lambda}|\boldsymbol{w}\right) \tag{5}$$

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and $\boldsymbol{q} = \{q_t\}_{t=1}^{T}$ is one particular state sequence. To simplify the equations in the following, we drop the sequence space $\mathcal{Q}$.

Since the logarithm is a monotonically increasing function, we define the equivalent optimization problem

$$\hat{\boldsymbol{\lambda}} = \arg\max_{\boldsymbol{\lambda}} \left\{ \log \left(\sum_{\boldsymbol{q}} f_{\boldsymbol{q}}(\boldsymbol{\lambda}) \right) \right\} \tag{6}$$

$$= \arg\max_{\boldsymbol{\lambda}} \left\{ \mathcal{F}(\boldsymbol{\lambda}) \right\}, \tag{7}$$

with the new objective function

$$\mathcal{F}(\boldsymbol{\lambda}) = \log \left(\sum_{\boldsymbol{q}} f_{\boldsymbol{q}}(\boldsymbol{\lambda}) \right). \tag{8}$$

Following the technique in [25], it can be shown[1] that

$$Q(\boldsymbol{\lambda}, \boldsymbol{\lambda}') = \sum_{\boldsymbol{q}} \frac{f_{\boldsymbol{q}}(\lambda')}{\sum_{\overline{\boldsymbol{q}}} f_{\overline{\boldsymbol{q}}}(\boldsymbol{\lambda}')} \log \left(f_{\boldsymbol{q}}(\boldsymbol{\lambda}) \right) \tag{9}$$

is a strong-sense auxiliary function of the objective function $\mathcal{F}(\boldsymbol{\lambda})$ in (8). This means that by iteratively maximizing $Q(\boldsymbol{\lambda}, \boldsymbol{\lambda}')$, a local maximum $\hat{\boldsymbol{\lambda}}$ of $\mathcal{F}(\boldsymbol{\lambda})$ can be found. In (9), $\overline{\boldsymbol{q}}$ is a summation index over all state sequences.

The joint log-probability $\log(f_{\boldsymbol{q}}(\boldsymbol{\lambda}))$ in (9) can be factored as:

$$\log(f_{\boldsymbol{q}}(\boldsymbol{\lambda})) = \log \left(\mathrm{p}\left(\boldsymbol{O}, \boldsymbol{q} | \boldsymbol{w}, \boldsymbol{\lambda}\right) \mathrm{p}(\boldsymbol{\lambda} | \boldsymbol{w}) \right) \tag{10}$$

$$= \log \left(\pi_{q_0} \prod_{t=1}^{T} a_{q_{t-1}, q_t} b_{q_t, \lambda_t}(O_t) \mathrm{p}(\lambda_t) \right), \tag{11}$$

where π_{q_0} is the initial state probability. In (11), a_{q_{t-1}, q_t} is the transition probability from the composite state q_{t-1} to the composite state q_t and $b_{q_t, \lambda_t}(O_t)$ is the emission score of the composite state q_t given the audio-visual observation vector O_t and the stream weight λ_t at time frame t. The step from (10) to (11) follows from the assumption

$$\mathrm{p}(\boldsymbol{\lambda} | \boldsymbol{w}) = \mathrm{p}(\boldsymbol{\lambda}) = \prod_{t=1}^{T} \mathrm{p}(\lambda_t). \tag{12}$$

In (12), the vector $\boldsymbol{\lambda}$ is assumed to be independent of the word sequence $\boldsymbol{w}$ and consists of independent and identically distributed (iid) random variables λ_t. Inserting (11) in (9), we get

$$Q(\boldsymbol{\lambda}, \boldsymbol{\lambda}') = \sum_{\boldsymbol{q}} \frac{f_{\boldsymbol{q}}(\boldsymbol{\lambda}')}{\sum_{\overline{\boldsymbol{q}}} f_{\overline{\boldsymbol{q}}}(\boldsymbol{\lambda}')} \sum_{t=1}^{T} \log \left(b_{q_t, \lambda_t}(O_t) \mathrm{p}(\lambda_t) \right) + \underbrace{\sum_{\boldsymbol{q}} \frac{f_{\boldsymbol{q}}(\boldsymbol{\lambda}')}{\sum_{\overline{\boldsymbol{q}}} f_{\overline{\boldsymbol{q}}}(\boldsymbol{\lambda}')} \log \left(\pi_{q_0} \prod_{t=1}^{T} a_{q_{t-1}, q_t} \right)}_{\text{independent of } \boldsymbol{\lambda}}. \tag{13}$$

Neglecting the terms independent of $\boldsymbol{\lambda}$ in (13), we get

$$Q(\boldsymbol{\lambda}, \boldsymbol{\lambda}') = \sum_{\boldsymbol{q}} \sum_{t=1}^{T} \frac{f_{\boldsymbol{q}}(\boldsymbol{\lambda}')}{\sum_{\overline{\boldsymbol{q}}} f_{\overline{\boldsymbol{q}}}(\boldsymbol{\lambda}')} \log \left(b_{q_t, \lambda_t}(O_t) \mathrm{p}(\lambda_t) \right) \tag{14}$$

$$= \sum_{t=1}^{T} \sum_{i=(1,1)}^{(N_A, N_V)} \gamma'_t(i) \log \left(b_{q_t=i, \lambda_t}(O_t) \mathrm{p}(\lambda_t) \right) \tag{15}$$

$$= \sum_{t=1}^{T} Q(\lambda_t, \boldsymbol{\lambda}'), \tag{16}$$

where

$$\gamma'_t(i) = \frac{f_{q_t=i}(\boldsymbol{\lambda}')}{\sum_{j=(1,1)}^{(N_A, N_V)} f_{q_t=j}(\boldsymbol{\lambda}')}, \tag{17}$$

and

$$f_{q_t=i}(\boldsymbol{\lambda}') = \mathrm{p}\left(O_t, q_t = i, \boldsymbol{\lambda}' | \boldsymbol{w}\right). \tag{18}$$

The coupled (composite) state occupation probability $\gamma'_t(i)$ is the probability of being in the coupled state i at time frame t. The auxiliary function in (15) can now be optimized by separately optimizing $Q(\lambda_t, \boldsymbol{\lambda}')$ for each frame t. By applying (1) to (15) we get the following expression for the auxiliary function at each t:

$$Q(\lambda_t, \boldsymbol{\lambda}') = \sum_{i=(1,1)}^{(N_A, N_V)} \gamma'_t(i) \log \left(b_{q_t=i, \lambda_t}(O_t) \right) + \sum_{i=(1,1)}^{(N_A, N_V)} \gamma'_t(i) \log \left(\mathrm{p}(\lambda_t) \right) \tag{19}$$

$$= \left[\sum_{i=(1,1)}^{(N_A, N_V)} \gamma'_t(i) \log \left(\frac{\mathrm{p}(O_{At} | q_{At} = i_A)}{\mathrm{p}(O_{Vt} | q_{Vt} = i_V)} \right) \right] \lambda_t + \underbrace{\sum_{i=(1,1)}^{(N_A, N_V)} \gamma'_t(i) \log \mathrm{p}(O_{Vt} | q_{Vt})}_{\text{independent of } \lambda_t} + \log \mathrm{p}(\lambda_t) \sum_{i=(1,1)}^{(N_A, N_V)} \gamma'_t(i), \tag{20}$$

where i_A and i_V are the single-modality acoustical and visual states composing the coupled state $i = (i_A, i_V)$. Neglecting the independent terms in (20) and using the fact that $\sum_i \gamma'_t(i) = 1$, we get

$$Q(\lambda_t, \boldsymbol{\lambda}') = \left[\sum_{i=(1,1)}^{(N_A, N_V)} \gamma'_t(i) \log \left(\frac{\mathrm{p}(O_{At} | q_{At} = i_A)}{\mathrm{p}(O_{Vt} | q_{Vt} = i_V)} \right) \right] \lambda_t + \log \mathrm{p}(\lambda_t). \tag{21}$$

If λ_t is assumed to be uniformly distributed, the auxiliary function (21) will be a linear function of λ_t. Optimizing this function leads to the problem already reported in [26], [27], that λ_t can take only the boundary values, i.e., $\lambda_t \in \{0, 1\}$. In other words, at each time frame t, one stream should be turned off. Which stream is unnecessary depends only on the sign—not on

[1]See Appendix A

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the magnitude—of the derivative of (21). However, if λ_t is assumed to be normally distributed with mean μ_λ and standard deviation σ_λ, the auxiliary function will be a quadratic function and we need to find

$$\hat{\lambda}_t = \arg\max_{\lambda_t} \left\{ \left[\sum_{i=(1,1)}^{(N_A,N_V)} \gamma'_t(i) \log \left(\frac{\mathrm{p}(O_{At}|q_{At}=i_A)}{\mathrm{p}(O_{Vt}|q_{Vt}=i_V)} \right) \right] \lambda_t - 0.5 \frac{(\lambda_t - \mu_\lambda)^2}{\sigma_\lambda^2} \right\} \quad \text{s.t.} \quad 0 \le \lambda_t \le 1. \tag{22}$$

The solution of this problem can be expressed as follows:

$$\hat{\lambda}_t = \begin{cases} \hat{\lambda}_{t_u} & \text{for} \quad 0 \le \hat{\lambda}_{t_u} \le 1 \\ 1 & \text{for} \quad \hat{\lambda}_{t_u} > 1 \\ 0 & \text{for} \quad \hat{\lambda}_{t_u} < 0, \end{cases} \tag{23}$$

where

$$\hat{\lambda}_{t_u} = \mu_\lambda + \sigma_\lambda^2 \sum_{i=(1,1)}^{(N_A,N_V)} \gamma'_t(i) \log \left(\frac{\mathrm{p}(O_{At}|q_{At}=i_A)}{\mathrm{p}(O_{Vt}|q_{Vt}=i_V)} \right) \tag{24}$$

is the solution of (22), but without constraints. Equations (23) and (24) represent the maximization step of the EM algorithm. In order to compute (24), the composite state occupation probabilities $\gamma'_t(i)$ should be found for each composite state at all time frames. As can be seen in (17), estimating $\gamma'_t(i)$, which represents the expectation step of the EM algorithm, requires a predefined initial stream weight vector $\boldsymbol{\lambda}'$. Finding a reasonable starting vector $\boldsymbol{\lambda}'$ is a critical issue as the EM algorithm only estimates a local maximum near this starting vector. In the next section we propose a greedy layer-wise optimization algorithm to suitably find an initialization vector $\boldsymbol{\lambda}'$.

B. EM Initialization

In order to find an initial vector $\boldsymbol{\lambda}'$, we apply a layer-wise greedy optimization algorithm as follows: At time frame t, we estimate λ'_t as

$$\lambda'_t = \arg\max_{\lambda_t} \left\{ \mathrm{p}\left(\boldsymbol{O}_{1,\cdots,t}, \lambda_t | \boldsymbol{\lambda'}_{1,\cdots,t-1}, \boldsymbol{w} \right) \right\} \tag{25}$$

$$= \arg\max_{\lambda_t} \left\{ \mathcal{F}_{\text{init}}\left(\lambda_t\right) \right\}. \tag{26}$$

The objective function $\mathcal{F}_{\text{init}}(\lambda_t)$ can be reformulated as follows:

$$\mathcal{F}_{\text{init}}\left(\lambda_t\right) = \mathrm{p}(\boldsymbol{O}_{1\cdots t}|\lambda_t, \boldsymbol{\lambda'}_{1\cdots t-1}, \boldsymbol{w}) \mathrm{p}(\lambda_t|\boldsymbol{\lambda'}_{1\cdots t-1}, \boldsymbol{w}) \tag{27}$$

$$= \sum_{i=(1,1)}^{(N_A,N_V)} \mathrm{p}(\boldsymbol{O}_{1\cdots t}, q_t = i|\lambda_t, \boldsymbol{\lambda'}_{1\cdots t-1}, \boldsymbol{w}) \mathrm{p}\left(\lambda_t\right) \tag{28}$$

$$= \mathrm{p}\left(\lambda_t\right) \sum_{i=(1,1)}^{(N_A,N_V)} \omega_t(i) b_{q_t,\lambda_t}(O_t). \tag{29}$$

The weights $\omega_t(i)$ in (29) can be written as

$$\omega_t(i) = \begin{cases} \pi_i & \text{for} \quad t = 1, \\ \sum_{j=(1,1)}^{(N_A,N_V)} a_{i,j} \alpha'_{t-1}(j) & \text{otherwise}, \end{cases} \tag{30}$$

where $\alpha'_{t-1}(j)$ is the joint probability of the partial observation sequence $\boldsymbol{O}_{1\cdots t-1}$ and state $q_{t-1} = j$ given the correct word sequence $\boldsymbol{w}$, i.e.,

$$\alpha'_{t-1}(j) = \mathrm{p}\left(\boldsymbol{O}_{1\cdots t-1}, q_{t-1} = j|\boldsymbol{\lambda'}_{1\cdots t-1}, \boldsymbol{w}\right). \tag{31}$$

The values of the forward parameters α' are computed using the forward algorithm [25] and the state emission scores defined in (1), in turn, are computed using the optimized stream weights of the preceding time frames $\boldsymbol{\lambda'}_{1\cdots t-1}$. Applying (1) to (29), we obtain the final expression for the objective function:

$$\mathcal{F}_{\text{init}}\left(\lambda_t\right) = \sum_{i=(1,1)}^{(N_A,N_V)} \omega_t(i) \left(\frac{\mathrm{p}(O_{A_t}|q_t=i)}{\mathrm{p}(O_{V_t}|q_t=i)} \right)^{\lambda_t} \mathrm{p}(O_{V_t}|q_t=i) \times \mathrm{p}\left(\lambda_t\right) \quad \text{s.t.} \quad 0 \le \lambda_t \le 1. \tag{32}$$

Assuming a Gaussian prior, i.e., $\mathrm{p}(\lambda_t) = \mathcal{N}(\lambda_t, \mu_\lambda, \sigma_\lambda^2)$, as discussed in Section IV-A, the objective functions in (32) are convex in the feasible region, i.e. $0 \le \lambda_t \le 1$, and can be optimized by gradient ascent [28]. The inequality constraints can be taken into account by adding a logarithmic barrier function.

C. Summary

The last piece of the puzzle is how to set the prior parameters μ_λ and σ_λ. A plausible choice of the mean μ_λ, which we call the bias parameter, is the global fixed stream weight λ_G, which can be found experimentally by grid search [11]. We consider the standard deviation σ_λ as the sensitivity parameter because it controls the dynamic range of the stream weights. In order to find an optimal value for the sensitivity parameter σ_λ, we start with a small value, i.e. low dynamic range, and increase it iteratively until the estimated stream weight vector $\hat{\boldsymbol{\lambda}}$ contains only binary values. Indeed, increasing σ_λ more than that does not make any sense because it will not change the resulting $\hat{\boldsymbol{\lambda}}$. After each iteration, the utterance is blindly recognized using the estimated weights. Finally, we choose the weight vector $\hat{\boldsymbol{\lambda}}$ that achieves best recognition accuracy. The entire approach is summarized in Algorithm 1.

Fig. 2 shows one example of the oracle dynamic stream weights (ODSWs) estimated using this algorithm. It can be seen that the absolute values of the oracle stream weights decrease with decreasing SNRs, which is plausible. For different SNRs, however, the temporal evolution of the oracle stream weights is quite similar. It can be seen that the stream weights drop during silence intervals for all SNRs. The stream weights for silence intervals at high SNRs still take on smaller values than non-silence frames at low SNRs. A similar observation regarding the reliability measures has been reported in [19]: The reliability measure ratios in speech intervals are different from these ratios in silence intervals. It has also been shown in [7] that under adverse acoustical conditions, the modality that can better detect silence acquires a larger weight. This observation has led the authors of [19] to train different mapping functions for speech and silence intervals. Apart from these effects, it is difficult to recognize the correlation between the spectrogram of the speech signal and the oracle stream weights by inspection. This can be attributed to the fact that the stream

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Algorithm 1 Oracle Dynamic Stream Weights For CHMM-based AVASR

A. Set the prior parameters
(4) Set μ_λ to the global fixed stream weight λ_G.
(5) Initialize σ_λ with a small value, e.g., 0.1.

B. Initialization
(6) Find $\hat{\boldsymbol{\lambda}} = \boldsymbol{\lambda}'$ using the greedy layer-wise algorithm introduced in Section IV-B.

C. EM Algorithm
(1) Calculate $P = \mathrm{p}(\boldsymbol{O}, \hat{\boldsymbol{\lambda}}|\boldsymbol{w})$

E step
(1) Use $\hat{\boldsymbol{\lambda}}$ to calculate $\gamma_t'(i)$ for all time frames and all coupled states (a 3-D Matrix).

M step
(1) Update $\hat{\boldsymbol{\lambda}}$ using the procedure introduced in Section IV-A.

Convergence test
(1) Calculate $P^* = \mathrm{p}(\boldsymbol{O}, \hat{\boldsymbol{\lambda}}|\boldsymbol{w})$
(2) If $(P^* - P > \epsilon)$
$P = P^*$, go to (5)
else
go to (9).

D. Recognition
(3) Use the estimated $\hat{\boldsymbol{\lambda}}$ to blindly recognize the training utterance and calculate the accuracy $A(\sigma_\lambda)$.

E. Iteration
(4) Increment σ_λ and repeat (3-10) until all values of $\hat{\boldsymbol{\lambda}}$ become zero or one.
(5) Choose $\hat{\boldsymbol{\lambda}}$ that achieves maximum accuracy.

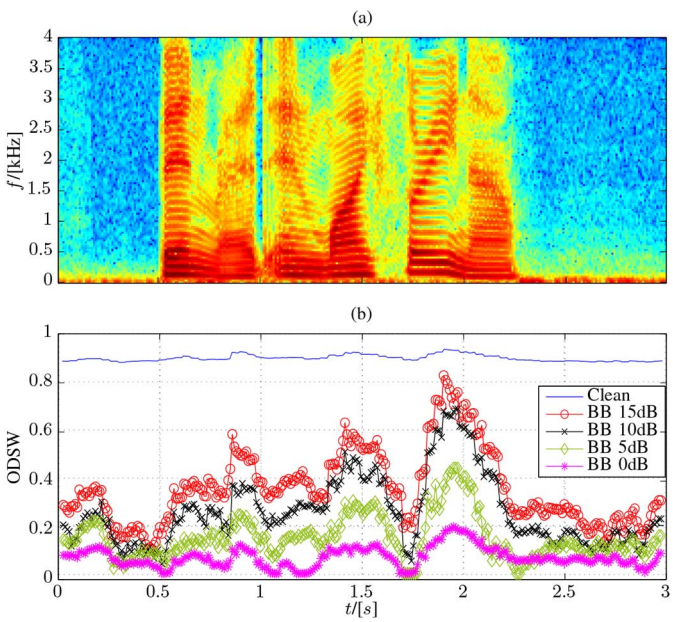


Fig. 2. (a) Spectrogram of a clean signal. (b) EM-based oracle dynamic stream weights (ODSWs) for the corresponding clean and distorted signals. The noise applied here is babble noise with the SNR $\in \{15, 10, 5, 0\}$ dB.

weights do not only depend on the acoustical environment, but also on the temporal information content of one modality compared to the other.

V. Reliability Measures

Reliability measures can be grouped into two main categories: signal-based and model-based reliability measures. The most important acoustical signal-based reliability measure is the signal-to-noise ratio (SNR). In this paper, we use a Wiener filter [29] to enhance the audio signal before extracting the acoustical features. The noise power estimate needed for the Wiener filter is obtained using improved minima controlled recursive averaging (IMCRA) [30],–[32]. The clean signal and noise estimated by this speech pre-processor can be used to estimate the instantaneous *a priori* SNRs as the ratio of the estimated clean signal energy $S(t)$ to the estimated noise energy $N(t)$ at time frame t:

$$\mathrm{SNR}(t) = 10 \log \left(\frac{S(t)}{N(t)} \right). \tag{33}$$

The instantaneous SNR measures the absolute confidence of the acoustical modality at each time frame. The SNRs, however, do not reflect the degree to which the acoustical features are matched to their model, e.g., in the case of multi-condition training. Moreover, the SNRs do not measure the ratio of confidences of both modalities. Therefore, using the SNR as the only reliability measure implicitly assumes that the visual modality has a constant reliability, which is not always true. This explains the need for additional reliability measures such as the memory-less instantaneous entropy H [33],–[35] and dispersion D [4], [7], [19].

The instantaneous entropy can be computed independently for each modality by

$$\mathrm{H}_s(t) = -\sum_{i=1}^{N_s} \mathrm{p}\left(q_s(t) = i|O_s(t)\right) \log\left(\mathrm{p}\left(q_s(t) = i|O_s(t)\right)\right). \tag{34}$$

In (34), $\mathrm{p}(q_s(t) = i|O_s(t))$ is the instantaneous probability distribution of the single-modality states q_s, $s \in \{A, V\}$, given the single-modality feature vector O_s at time frame t. These posteriors can be computed via

$$\mathrm{p}\left(q_s(t) = i|O_s(t)\right) = \frac{b_{q_s(t)=i}(O_s(t))\mathrm{p}\left(q_s(t) = i\right)}{\sum_{j=1}^{N_s} b_{q_s(t)=j}(O_s(t))\mathrm{p}\left(q_s(t) = j\right)}. \tag{35}$$

The prior $\mathrm{p}(q_s(t) = i)$ can be found during the training procedure of the single-modality marginal models as follows:

$$\mathrm{p}\left(q_s = i\right) = \frac{1}{R} \sum_{f=1}^{F} \sum_{t=1}^{T_f} \mathrm{p}\left(q_s(t) = i|\boldsymbol{w}_f, \boldsymbol{O}_{sf}\right). \tag{36}$$

In (36), f is a summation index over the utterances used in training and T_f denotes the length of each utterance. Given an observation sequence $\boldsymbol{O}_{sf}$ and its corresponding correctly-labeled word sequence $\boldsymbol{w}_f$, whose models contain the state $q_s = i$, the posterior probabilities $\mathrm{p}(q_s(t) = i|\boldsymbol{w}_f, \boldsymbol{O}_f)$ can be found from the last estimation step of the Baum-Welch algorithm [25] used for marginal-model training of the modality s. The summation in (36) is applied over all word sequences $\boldsymbol{w}_f$ that contain

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the state $q_s = i$, and R is a normalization constant that can be computed via

$$R = \sum_{i=1}^{N_s} \sum_{f=1}^{F} \sum_{t=1}^{T_f} \mathrm{p}\left(q_s(t) = i | \boldsymbol{w}_f, \boldsymbol{O}_{sf}\right). \tag{37}$$

Assuming that $q_s(t)$ is a discrete random variable, the entropy computed in (34) is a measure of the uncertainty in this random variable that remains after knowing the observation O_s at time frame t. From this definition, the smaller the value of the entropy in (34) becomes, the more useful the observation is at this time frame.

The second model-based reliability measure is the dispersion, which can be computed as follows:

$$\mathrm{D}_s(t) = \frac{2}{K(K-1)} \sum_{l=1}^{K} \sum_{m=l+1}^{K} \log \frac{\mathrm{p}\left(q_s(t) = l | O_s(t)\right)}{\mathrm{p}\left(q_s(t) = m | O_s(t)\right)}. \tag{38}$$

Before the dispersion is calculated as in (38), the posteriors need to be arranged in descending order. Unlike for the entropy, where the summation is applied over all model states N_s, for the dispersion, the summation can be limited to the K largest posteriors. It is worth mentioning that the dispersion is alternatively defined in [19] using likelihoods $\mathrm{p}(O_s(t)|q_s(t) = l)$ instead of posteriors.

The dispersion is a heuristic measure of the decoder's discriminative power: The larger the dispersion, the less confused is the decoder and the larger is its capability of identifying the best state at frame t.

We call the entropy and the dispersion memory-less measures, because they do not take into account past information such as the previous states and their transition probabilities to the current state. Therefore, even posteriors of states that are unreachable at time because of the model topology constraints are still involved in the summations of Equation (34) and (38). However, replacing the memory-less posteriors in (34) and (38) by the full posteriors $\mathrm{p}(q_s(t) = l|\boldsymbol{O}_s(1 \ldots t))$ reduces the overall performance. Therefore, the memory-less reliability measures defined in Equation (34) and (38) will be used in the following.

In addition to the five conventional reliability measures mentioned above, namely, the SNR and the audio and video dispersions and entropies, we propose to use hard (binary) and soft (probabilistic) voice activity detection (VAD) cues as signal-based reliability measures. The VAD cues are involved to allow for the mapping function to behave differently in silence and speech intervals.

Finally, we propose to consider the so-called acoustical feature uncertainties [2] as an additional signal-based reliability measure. The feature uncertainties can be estimated when speech signals are pre-processed by a speech enhancement algorithm, e.g. to reduce signal distortions due to additive and convolutive noise. The enhanced signals are not compensated perfectly and typically contain residual noise, estimation errors, and even artifacts introduced by the pre-processors themselves. Fortunately, Bayesian speech enhancement algorithms like the Wiener filter—the speech enhancement algorithm used in this study—can compute the uncertainty of their estimation.

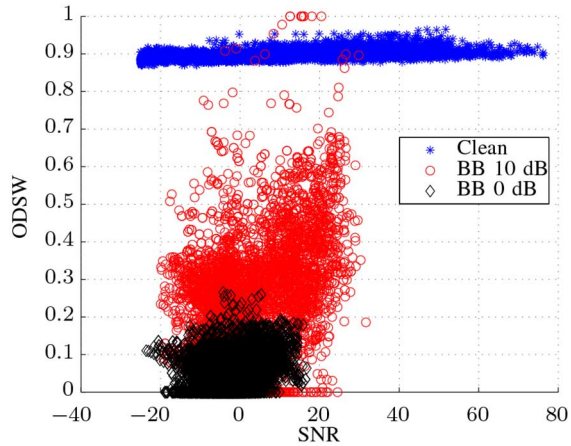


Fig. 3. Oracle dynamic stream weights (ODSWs) versus frame-based SNRs. Data points extracted from clean data and from signals embedded in babble (BB) noise at 10 dB and 0 dB are shown.

Particularly speaking, as the Wiener filter computes a point estimate $\hat{x}_{t,f}$ of the clean discrete Fourier transform (DFT) coefficient $x_{t,f}$ at each time frame t and frequency bin f as

$$\hat{x}_{t,f} = \mathrm{E}\left[x_{t,f} | y_{t,f}\right], \tag{39}$$

the variance $\sigma^2 = \mathrm{Var}[x_{t,f}|y_{t,f}]$ can be considered as the uncertainty of the estimated features $\hat{x}_{t,f}$[36]. In (39), $y_{t,f}$ is the corresponding observed noisy discrete Fourier transform (DFT) coefficient.

All mentioned signal-based and model-based reliability measures are concatenated to form multidimensional feature vectors that are used for training the mapping function parameters and later for blind estimation of dynamic stream weights.

VI. Mapping Function for Stream Weight Estimation

The last sub-problem of the stream weight regression problem introduced in Section I is choosing a suitable function to map the reliability measure features to stream weights. To get an insight into which type of function could be used, we plot frame-based SNRs, dispersions and entropies against the corresponding oracle dynamic stream weights provided by the proposed EM algorithm in Figs. 3, 4, and 5. The data points in these figures have been extracted from utterances arbitrarily selected from a clean dataset and from a dataset artificially distorted with babble noise at average SNRs of 0 and 10 dB.

Figs. 3–5 show that the scatter plots of all reliability measures against stream weights look fairly chaotic. According to these plots, there is actually no direct functional relationship between the SNR, the dispersion or the entropy and the stream weight. This shows the need for deploying more sophisticated mapping functions that accept multidimensional input feature vectors. Therefore, we propose using an MLP or a multidimensional logistic function as this mapping function.

VII. The Entire Framework at a Glance

In Fig. 6, we summarize the entire framework proposed here for the blind estimation of dynamic stream weights. In the training phase, the EM algorithm uses the audio and visual

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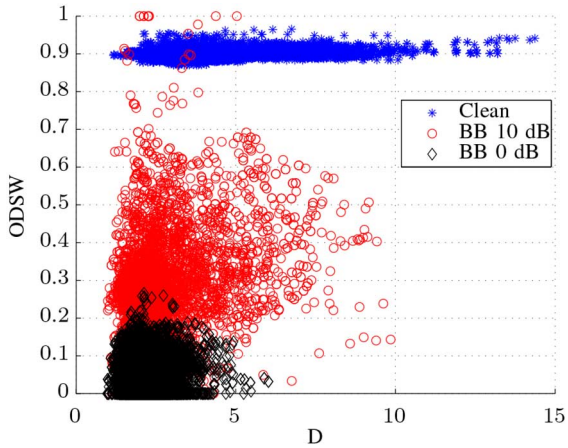


Fig. 4. Oracle dynamic stream weights versus frame-based dispersions.

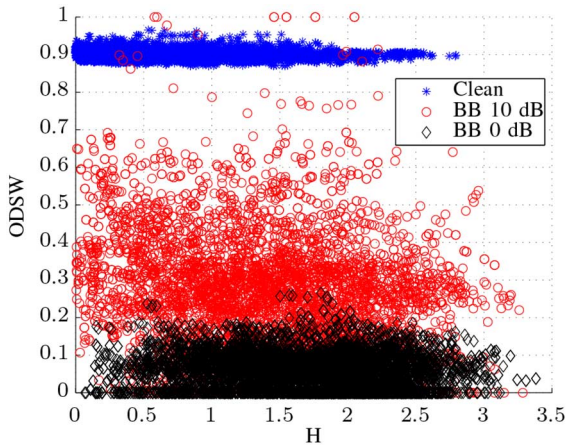


Fig. 5. Oracle dynamic stream weights versus frame-based entropies.

features of the development set in conjunction with their corresponding labels to estimate the oracle dynamic stream weights as discussed in Section IV. These oracle dynamic stream weights are then used with all reliability measure features discussed in Section V to train the parameters of the chosen mapping function (an MLP or a multidimensional logistic function). During testing, the mapping function is used to map the reliability measure feature vectors of the test set to stream weights. Finally, the estimated stream weights are deployed in the CHMM-AVASR system using (1).

The framework shown in Fig. 6 can be applied in real-time due to the following aspects: During decoding, all reliability measures are computed from auxiliary information already computed by the speech pre-processor and by the log-likelihood calculator. Moreover, computing the stream weights through the neural network feed-forward process or by applying the logistic function can be done in real time on a standard PC. Finally, real-time implementation of CHMM-AV-ASR is possible, see, e.g. [37].

VIII. Experiments and Results

A. Experimental Setup

We use the Grid audio-visual corpus [38] as a small-vocabulary ASR task to evaluate all approaches. The Grid database consists of 34000 sentences spoken

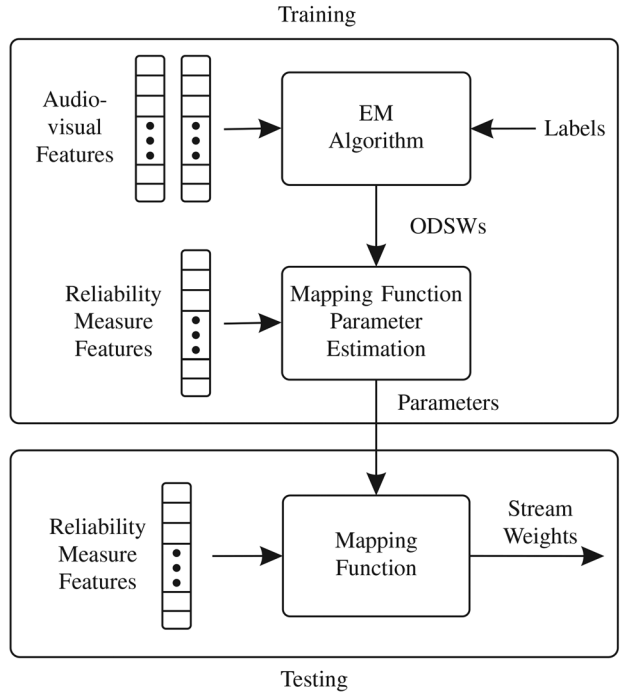


Fig. 6. Proposed framework for dynamic stream weight estimation.

by 34 speakers, i.e., 1000 sentences per speaker. Of these, we use 33 speakers, as the video data of one speaker is missing in the database. The task of the Grid corpus is to recognize sentences from a small vocabulary (51 words) with a fixed grammar of the form: `command - color - preposition - letter - digit - adverb`.

The utterances of each speaker have been divided into a training (90%), a development (5%) and a test set (5%). In order to create corrupted versions of the Grid corpus, two acoustical noise types have been artificially added to the signals of the test and development sets at different SNRs in the range from 15 dB down to 0 dB. We have chosen speech babble and white noise as examples of non-stationary and stationary noise, respectively. The noise signals have been taken from the NOISEX database [39].

All signals have been down-sampled to $f_s = 8$ kHz and enhanced using a Wiener filter. The noise power estimate needed for the Wiener filter has been obtained using improved minima-controlled recursive averaging (IMCRA). We have used the first 13 static MFCCs [40] along with the 26 corresponding Δ and $\Delta\Delta$ coefficients as the audio features. The static MFCC features have been extracted according to the ETSI advanced front-end (AFE) recommendations [41] as follows: The spectrograms have been computed using a 25 ms Hanning window with 15 ms overlap and an FFT length of 256 samples. A 24-channel mel filterbank has been applied to the magnitude-squared spectrogram followed by the DCT. The video features are the first 64 DCT coefficients, encoding the appearance and shape of the speaker's mouth. The corresponding mouth region was determined automatically by a Viola-Jones face and mouth detector [42]. The dimension of both audio and video features has been reduced to 31 by linear discriminant analysis (LDA) [43], [44].

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TABLE I
IDEAL GLOBAL STREAM WEIGHTS AND THE CORRESPONDING ACCURACIES

Noise Type	SNR [dB]	λ_G^{ideal}	Audio-visual Accuracy
Babble	15	0.4	94.29
	10	0.3	90.93
	5	0.2	87.34
	0	0.1	84.83
White	15	0.5	94.22
	10	0.3	91.13
	5	0.2	88.41
	0	0.2	85.94
Clean	-	0.9	99.30

The marginal single-stream HMMs have been trained generatively using the clean signals of the training sets. Each HMM set consists of 51 whole-word HMMs and one silence HMM. The word HMMs are speaker-dependent left-to-right linear models, with three states per phone in audio HMMs and one state per phone in video HMMs. The audio and video state conditional probabilities are 3- and 4-component diagonal covariance GMMs, respectively. The Gaussian mixtures have been split in the direction of the first eigenvector of the data covariance. This strategy has proven useful in [45], [46]. All experiments have been conducted using our in-house Java Audio-visual SPEech Recognizer Jasper, see, e.g., [47].

The ASR performance has been evaluated in terms of word accuracy, which is defined as a function of the number of reference labels (N), substitutions (S), insertions (I) and deletions (D) via

$$\%Accuracy = \frac{N - D - S - I}{N} \cdot 100. \tag{40}$$

B. Oracle Results

In this section, we introduce the oracle results, which have been obtained using the correct transcriptions of the utterances. The visual ASR accuracy in all experiments is 83.54% and the results are obtained using the development set.

Table I shows the oracle results, which are obtained using the ideal global stream weights λ_G^{ideal}. The stream weights shown in this table are found using grid search.

Table II shows the word accuracies of the audio-only ASR and the AVASR when the following oracle stream weights are used:

- the ideal global stream weight,
- the initialized oracle dynamic stream weights, which are estimated as discussed in Section IV-B,
- the EM-based oracle dynamic stream weights with random initialization, and
- the EM-based oracle dynamic stream weights introduced in Algorithm 1.

Using the proposed EM-based oracle dynamic stream weights improves the recognition performance of the AVASR in every considered case, by on average 3.17% compared to the ideal global stream weights. The results with asterisks achieved statistically significant improvements relative to the results obtained using the best global stream weights, according to Fisher's exact test [48] applied at $\mathrm{p} = 0.01$.

TABLE II
ASR PERFORMANCE IN DIFFERENT ENVIRONMENTS. THE AVASR RESULTS ARE OBTAINED USING THE IDEAL GLOBAL STREAM WEIGHTS, THE INITIAL STREAM WEIGHTS, THE RANDOMLY INITIALIZED ORACLE DYNAMIC STREAM WEIGHTS AND THE ORACLE DYNAMIC STREAM WEIGHTS

Noise Type	SNR [dB]	Audio only	Audio-visual			
			λ_G^{ideal}	Initial	ODSW-R	ODSW
Babble	15	84.97	94.29	95.27	95.70	96.77 $*$
	10	68.79	90.93	92.32	93.02	94.82 $*$
	5	47.35	87.34	88.21	88.97	91.73 $*$
	0	30.82	84.83	85.29	85.92	88.92 $*$
White	15	84.79	94.22	95.07	95.59	96.72 $*$
	10	68.59	91.13	91.94	92.59	94.52 $*$
	5	51.70	88.41	88.81	88.97	91.83 $*$
	0	36.23	85.94	86.46	87.46	90.23 $*$
Clean	-	99.15	99.30	99.31	99.33	99.41
Avg.	-	63.60	90.71	91.41	91,95	93.88 $*$

Comparing the results of the third and the fifth column in Table II, it can be seen that the EM iterations significantly improve the results obtained from the proposed initialization approach (Initial) discussed in Section IV-B. Indeed, estimating new stream weights in each iteration also changes the frame-state alignment. This in turn yields a new set of stream weights and so on. The results show how critical these EM iterations are for improving the estimation of the stream weights.

It is well known that the EM algorithm converges only to a local maximum. Therefore, the solution of the optimization problem differs according to the starting point. This can be seen by comparing the results obtained using the EM algorithm with random initialization (ODSW-R) and with the Viterbi-like initialization (ODSW) introduced in Section IV-B.

Finally, Table II shows the large improvements that can be achieved by optimally fusing the audio and video stream compared to only using a single stream.

C. Blind Estimation of Stream Weights Using Global Oracle Targets

In this section, we provide the baseline accuracies obtained using the approach introduced in [19] and summarized in Section II. In this approach, the ideal global stream weights λ_G^{ideal} that minimize the word error rate (WER) for a specific acoustic condition are used as training targets for univariate mapping functions. The ideal global stream weights are determined for each development dataset, corresponding to one specific SNR and noise type, using grid search. This means that we have only nine ideal global weights for nine development datasets, i.e. one clean dataset, four datasets distorted by babble noise and four distorted by white noise either with SNRs $\in \{15\,\text{dB}, 10\,\text{dB}, 5\,\text{dB}, 0\,\text{dB}\}$. Since the reliability measures introduced in Section V have been estimated per frame, they are averaged over all frames of the development set (298 frames/signal $\times$ 1549 signals) to find one corresponding input for each dataset. In this experiment, we use the audio-only entropies and dispersions and the SNRs as reliability measures, corresponding to the reference work in [19].

The functions that map reliability measures to stream weights are often chosen without giving explicit reasons for optimality. In [7] and [19], a sigmoid and a second-order exponential function have been used. In this paper, we compare the performance of five different mapping functions for each reliability measure.

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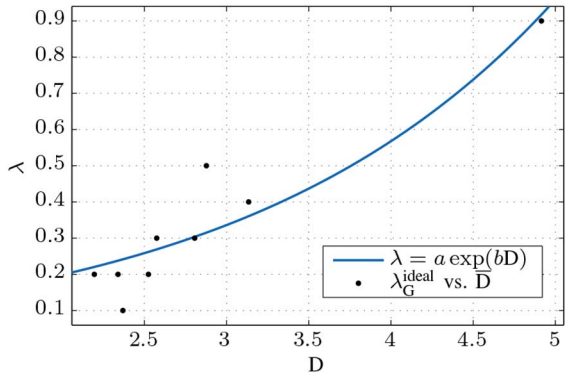


Fig. 7. First-order exponential function mapping the dispersion D to the stream weight λ. The black points are the global stream weights $\lambda_G^{\mathrm{ideal}}$ as a function of the averaged dispersions $\overline{\mathrm{D}}$ for each development set.

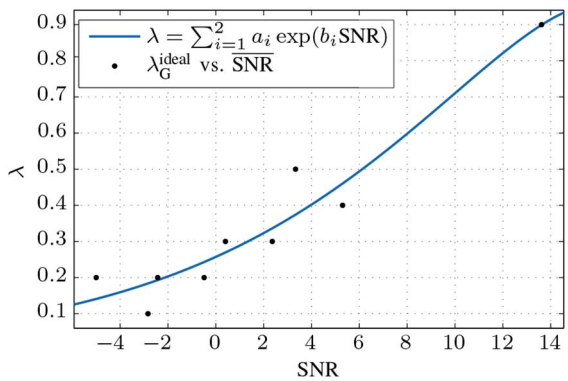


Fig. 8. Second-order exponential function mapping the SNR to the stream weight λ.

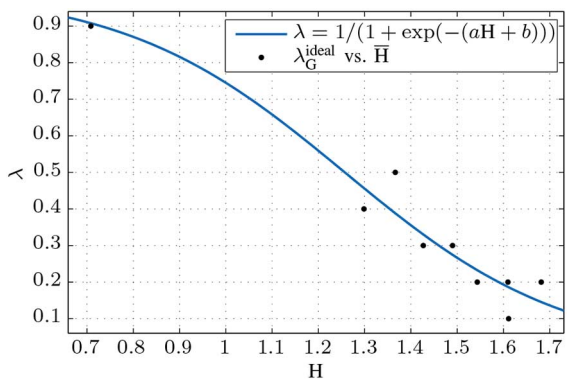


Fig. 9. Sigmoid function mapping the entropy H to the stream weight λ.

The chosen functions are first- and second-order exponential functions, first- and second-order polynomial functions (linear and quadratic functions) and a sigmoid function. The outputs of those mapping functions, whose range is not bounded between zero and one, are clipped to these values.

The mapping function parameters can be estimated using for example a minimum mean square error (MMSE) objective function [19]. Figs. 7, 8 and 9 show examples of the mapping functions estimated using the Matlab curve fitting toolbox.

During recognition, for each frame, the estimated mapping functions are employed to estimate the frame-dependent stream weights. The stream weights are then used in the likelihood evaluations according to Equation (1).

Tables III–VIII show the AVASR word accuracy, when the AVASR system is used to recognize the development and test set signals. The similar performance of this approach on both sets shows its good generalization to unseen data. However, it can be observed that there is no single mapping function that performs best for all reliability measures under all acoustical conditions. The first-order exponential function, however, gives the best average performance with the model-based reliability measures, i.e., entropy and dispersion. On the other hand, when the SNR is used, all mapping functions give almost the same average performance.

In very non-stationary noisy environments and at high noise levels, the performance of the SNR in general is significantly better than the model-based reliability measures, which points at the non-robustness of these measures under such conditions. On

TABLE III
DEVELOPMENT SET AVASR WORD ACCURACIES OBTAINED USING FRAME-DEPENDENT STREAM WEIGHTS, WHICH ARE MAPPED FROM THE ESTIMATED FRAME-DEPENDENT *DISPERSIONS* USING DIFFERENT UNIVARIATE MAPPING FUNCTIONS UNDER DIFFERENT ACOUSTICAL CONDITIONS

Noise Type	SNR [dB]	Poly. 1st	Poly. 2nd	exp. 1st	exp. 2nd	Sigm.
Babble	15	91.39	90.96	92.60	90.52	91.88
	10	86.70	86.10	88.47	85.88	87.00
	5	83.04	82.33	84.40	83.08	82.81
	0	78.43	77.95	78.84	79.55	78.11
White	15	91.35	90.96	92.53	90.21	91.82
	10	88.16	87.54	89.91	87.07	88.77
	5	85.46	85.14	87.13	84.96	86.27
	0	82.70	82.12	84.06	82.87	82.94
Clean	-	98.98	99.00	98.84	98.81	99.02
Avg.	-	87.36	86.90	88.53	86.99	87.62

TABLE IV
DEVELOPMENT SET AVASR WORD ACCURACIES OBTAINED USING FRAME-DEPENDENT STREAM WEIGHTS, WHICH ARE MAPPED FROM THE ESTIMATED FRAME-DEPENDENT *SNRS*

Noise Type	SNR [dB]	Poly. 1st	Poly. 2nd	exp. 1st	exp. 2nd	Sigm.
Babble	15	91.84	90.69	90.50	90.88	91.72
	10	87.97	87.95	87.77	88.35	88.12
	5	84.61	86.25	86.27	86.58	85.66
	0	83.00	81.69	83.83	83.28	83.76
White	15	90.02	88.92	88.72	88.97	90.22
	10	84.34	80.35	80.06	80.58	84.16
	5	81.40	81.04	80.84	82.09	81.58
	0	81.72	83.94	83.56	84.13	82.82
Clean	-	98.82	99.02	98.88	98.89	98.86
Avg.	-	87.08	86.65	86.71	87.08	87.43

TABLE V
DEVELOPMENT SET AVASR WORD ACCURACIES OBTAINED USING FRAME-DEPENDENT STREAM WEIGHTS, WHICH ARE MAPPED FROM THE ESTIMATED FRAME-DEPENDENT *ENTROPIES*

Noise Type	SNR [dB]	Poly. 1st	Poly. 2nd	exp. 1st	exp. 2nd	Sigm.
Babble	15	89.98	90.61	91.56	89.75	90.39
	10	84.59	83.48	86.24	84.58	84.78
	5	79.36	76.21	80.99	79.78	79.03
	0	73.98	68.58	74.40	74.99	72.65
White	15	89.89	90.74	91.55	89.90	90.31
	10	85.45	85.50	87.68	85.83	85.82
	5	82.70	81.71	84.68	83.10	83.12
	0	77.39	74.91	78.73	78.22	77.18
Clean	-	98.98	99.12	99.03	98.90	99.04
Avg.	-	84.70	83.43	86.10	85.01	84.70

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TABLE VI
TEST SET AVASR WORD ACCURACIES OBTAINED USING FRAME-DEPENDENT STREAM WEIGHTS, WHICH ARE MAPPED FROM THE ESTIMATED FRAME-DEPENDENT *DISPERSIONS*

Noise Type	SNR [dB]	Poly. 1st	Poly. 2nd	exp. 1st	exp. 2nd	Sigm.
Babble	15	91.31	91.06	92.05	90.98	91.51
	10	86.71	86.44	87.60	86.56	86.69
	5	82.13	81.40	83.53	82.49	81.70
	0	80.88	80.68	79.98	81.97	79.88
White	15	91.54	91.22	92.52	90.94	91.81
	10	87.64	87.29	89.15	86.97	87.99
	5	85.07	84.58	86.28	84.59	85.32
	0	83.46	83.24	84.29	84.09	83.57
Clean	-	98.74	98.74	98.79	98.73	98.74
Avg.	-	87.50	87.18	88.24	87.48	87.47

TABLE VII
TEST SET AVASR WORD ACCURACIES OBTAINED USING FRAME-DEPENDENT STREAM WEIGHTS, WHICH ARE MAPPED FROM THE ESTIMATED FRAME-DEPENDENT *SNRS*

Noise Type	SNR [dB]	Poly. 1st	Poly. 2nd	exp. 1st	exp. 2nd	Sigm.
Babble	15	90.87	89.40	89.24	89.36	90.82
	10	87.55	87.56	87.37	87.86	87.83
	5	83.14	84.52	84.47	85.00	84.26
	0	83.42	82.11	84.31	83.43	84.30
White	15	88.27	86.68	86.54	86.87	88.87
	10	82.52	78.60	78.42	78.63	82.23
	5	80.68	80.19	80.22	81.60	80.97
	0	81.73	83.68	83.61	84.02	83.05
Clean	-	98.60	98.69	98.59	98.60	98.63
Avg.	-	86.31	85.71	85.86	86.15	86.77

TABLE VIII
TEST SET AVASR WORD ACCURACIES OBTAINED USING FRAME-DEPENDENT STREAM WEIGHTS, WHICH ARE MAPPED FROM THE ESTIMATED FRAME-DEPENDENT *ENTROPIES*

Noise Type	SNR [dB]	Poly. 1st	Poly. 2nd	exp. 1st	exp. 2nd	Sigm.
Babble	15	90.38	90.28	91.59	90.51	90.64
	10	83.90	83.21	85.51	84.47	84.02
	5	77.18	73.91	79.05	78.29	76.80
	0	76.04	70.91	76.48	77.11	75.36
White	15	90.45	90.69	91.67	90.43	90.63
	10	85.31	84.76	87.13	85.69	85.26
	5	81.03	79.93	82.81	81.40	81.07
	0	79.73	77.33	81.22	80.47	79.68
Clean	-	98.88	98.91	98.85	98.83	98.88
Avg.	-	84.77	83.33	86.03	85.24	84.70

the other hand, in mild stationary noise, the performance of the SNR is worse than that of the model-based reliability measures. The above observations may change, however, when using another acoustical speech pre-processor for SNR estimation.

D. Blind Estimation of Stream Weights Using Oracle Dynamic Targets

In this section, we demonstrate the results obtained using the framework shown in Fig. 6 and compare them to the baseline results introduced in Section VIII-C. As mentioned previously, we use an MLP and a multivariate logistic function as alternative mapping functions. Training of these mapping functions has been done using the signals of all nine development datasets. From each signal, we have extracted a sequence of reliability measure feature vectors. The 31-dimensional reliability measure feature vectors are composed of the four model-based reliability measures, i.e., audio and video entropies and dispersions, and the 27 signal-based reliability measures, which contain the SNR, the hard and soft VAD cues, and the 24 acoustical feature uncertainties. The dimension of the uncertainties is reduced from the DFT length to 24 by linear transform using the Mel filterbank matrix. The binary and the probabilistic VAD features are provided in each time-frequency bin by the IMCRA approach [31], [32]. The average values of each of these two VAD feature vectors over all frequency bins are used in the reliability measure feature vector. The total number of the feature vectors extracted from the development set is about 1.15 million, which is large enough to well train the stream weight estimators. The same number of the oracle stream weights has been estimated using the EM algorithm discussed in Section IV.

The parameters of the logistic function have been estimated to minimize the regularized logistic regression cost function using the Polak-Ribiére conjugate gradient method [49].

We have tested different MLP architectures, changing the number of layers and the number of units in each layer. The experimental results show that increasing the depth of the MLP does not improve the overall performance of the AVASR system significantly. The architecture that has given the best performance on the development set is composed of an input layer with 31 units, two hidden layers each with 10 units and a one-dimensional output layer. In order to fairly compare deep architectures with conventional ones, we have initialized the MLP weights of different architectures using restricted-Boltzmann-machine (RBM) pre-training [50], [51]. Finally, we have fine-tuned all MLP weights via back-propagation [52], [53], with the reliability measure feature vectors as inputs and the oracle stream weights as target outputs. The Levenberg-Marquardt back-propagation algorithm of the Matlab neural network toolbox has been used for this purpose. The mean squared error between the MLP outputs and the oracle stream weights approaches 9.5×10^{-3} after 300 epochs.

Tables IX and X present the word accuracies of the AVASR when it is applied to the development set and the test set signals. It can be seen that again, no overfitting has occurred, as the performance on the development set is quite similar to that on the test set. In the last two columns of Tables IX and X, the stream weight estimator has been trained using the oracle dynamic stream weights that have been estimated using our proposed approach in Section IV. As discussed in Section VI, two types of mapping functions have been used, a logistic function and an MLP. The columns 3-5 in Tables IX and X show the best results of the baseline approach.

The results show that training stream weight estimators using oracle dynamic stream weights consistently and significantly improves the AVASR performance in every test condition compared to the baseline results, which are obtained using stream weight estimators trained on oracle global stream weights. Although the results obtained using the MLP are better than those using the logistic function in almost every case, the improvements are not statistically significant.

The results obtained using the proposed framework approach those obtained using the ideal global stream weights λ_G^{ideal}. Still,

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TABLE IX
DEVELOPMENT SET AVASR PERFORMANCE OBTAINED USING DYNAMIC STREAM WEIGHTS, ESTIMATED VIA LOGISTIC REGRESSION AND USING AN MLP

Noise Type	SNR dB	Global-weights-based			ODSW-based	
		Disp. exp. 1st	SNR Sigm.	Ent. exp. 1st	Logistic Curve Fit	MLP
Babble	15	92.60	91.72	91.56	94.02	**94.22**
	10	88.47	88.12	86.24	**90.65**	90.59
	5	84.40	85.66	80.99	86.69	**86.89**
	0	78.84	83.76	74.40	84.53	**84.99**
White	15	92.53	90.22	91.55	93.93	**94.09**
	10	89.91	84.16	87.68	90.82	**91.08**
	5	87.13	81.58	84.68	88.29	**88.35**
	0	84.06	82.82	78.73	85.56	**86.03**
Clean	-	98.84	98.86	99.03	99.05	**99.26**
Avg.	-	88.53	87.43	86.10	90.39	**90.61**

TABLE X
TEST SET AVASR PERFORMANCE OBTAINED USING DYNAMIC STREAM WEIGHTS, ESTIMATED VIA LOGISTIC REGRESSION AND USING AN MLP

Noise Type	SNR dB	Global-weights-based			ODSW-based	
		Disp. exp. 1st	SNR Sigm.	Ent. exp. 1st	Logistic Curve Fit	MLP
Babble	15	92.05	90.82	91.59	93.62	**93.74**
	10	87.60	87.83	85.51	90.31	**90.64**
	5	83.53	84.26	79.05	86.25	**86.99**
	0	79.98	84.30	76.48	84.95	**85.13**
White	15	92.52	88.87	91.67	93.62	**93.91**
	10	89.15	82.23	87.13	90.50	**90.87**
	5	86.28	80.97	82.81	87.70	**87.83**
	0	84.29	83.05	81.22	85.66	**85.85**
Clean	-	98.79	98.63	98.85	98.97	**99.07**
Avg.	-	88.24	86.77	86.03	90.17	**90.45**

there is a gap between the *estimated* stream weights and the *oracle* dynamic stream weights. We aim at reducing this gap between the achieved performance and this theoretical upper limit in future work, e.g. by finding additional—more robust—reliability measure features, and by including reliability measures concerning the quality of the video signal as well.

IX. CONCLUSIONS

Optimizing the contribution of the audio and video modality to the recognition decision using stream weights has a great impact on the overall performance of AVASR systems. In many prior works, the stream weights have been estimated using heuristically chosen mapping functions, which are trained in a supervised manner considering oracle global stream weights as their target outputs. In this paper, we have presented an EM algorithm that estimates oracle dynamic stream weights on a frame-by-frame basis for CHMMs. The EM algorithm solves the problem of pre-defining frame-state alignments and provides a new upper bound that can be achieved using AVASR systems. The numerous data points provided by the EM algorithm enable parameters of sophisticated multidimensional mapping functions such as MLPs and multidimensional logistic functions to be well trained. The presented framework enables all available reliability measures to be combined in high-dimensional input feature vectors and utilized together.

The AVASR results obtained using the proposed framework are significantly better than the baseline results in all considered test conditions. However, the results are still far from those obtained using EM-based oracle dynamic stream weights. These oracle weights are not usable in practice due to their reliance on the true word transcription, but can nonetheless serve as an indication of the maximum performance theoretically achievable by dynamic stream weighting.

We aim at extending each of the three parts of the introduced framework as follows: We will consider estimating the oracle dynamic stream weights using a discriminative objective function for the EM algorithm, as we believe that this will allow us to dispense with the assumption of certain priors of the PDF underlying the stream weights. Another future work direction, which can reduce the gap between the achieved performance and that obtained using oracle knowledge, is the inclusion of other noise-robust reliability measure features as well as video quality measures as additional inputs. Finally, we will consider improving the MLP performance by intermediately training it to detect high-level features like the noise type and level, and then using these features in conjunction with all other reliability measures to estimate the stream weights.

In this work, we have considered the performance of the proposed framework when the acoustical features are distorted by acoustical noise. Another aspect that needs to be considered in future work is the performance of the proposed framework under adverse visual conditions.

APPENDIX A
AUXILIARY FUNCTION

In order to prove that $Q(\boldsymbol{\lambda}, \boldsymbol{\lambda}')$ is an auxiliary function for $\mathcal{F}(\boldsymbol{\lambda})$, we need to prove the following inequality:

$$Q(\boldsymbol{\lambda}, \boldsymbol{\lambda}') - Q(\boldsymbol{\lambda}', \boldsymbol{\lambda}') \leq \mathcal{F}(\boldsymbol{\lambda}) - \mathcal{F}(\boldsymbol{\lambda}') \quad \forall \boldsymbol{\lambda}' \tag{41}$$

or

$$Q(\boldsymbol{\lambda}, \boldsymbol{\lambda}') - Q(\boldsymbol{\lambda}', \boldsymbol{\lambda}') - \mathcal{F}(\boldsymbol{\lambda}) + \mathcal{F}(\boldsymbol{\lambda}') \leq 0 \quad \forall \boldsymbol{\lambda}'. \tag{42}$$

For sake of simplicity, we use the following notations:

$$B'(\boldsymbol{q}) = \frac{f_{\boldsymbol{q}}(\boldsymbol{\lambda}')}{\sum_{\overline{\boldsymbol{q}}} f_{\overline{\boldsymbol{q}}}(\boldsymbol{\lambda}')} \quad \text{and} \quad B(\boldsymbol{q}) = \frac{f_{\boldsymbol{q}}(\boldsymbol{\lambda})}{\sum_{\overline{\boldsymbol{q}}} f_{\overline{\boldsymbol{q}}}(\boldsymbol{\lambda})}.$$

The auxiliary function can be rewritten as follows:

$$Q(\boldsymbol{\lambda}, \boldsymbol{\lambda}') = \sum_{\boldsymbol{q}} B'(\boldsymbol{q}) \log\left(f_{\boldsymbol{q}}(\boldsymbol{\lambda})\right) \tag{43}$$

$$= \sum_{\boldsymbol{q}} \Bigg[B'(\boldsymbol{q}) \Big(\log\left(f_{\boldsymbol{q}}(\boldsymbol{\lambda})\right) + \log\left(\sum_{\overline{\boldsymbol{q}}} f_{\overline{\boldsymbol{q}}}(\boldsymbol{\lambda})\right) - \log\left(\sum_{\overline{\boldsymbol{q}}} f_{\overline{\boldsymbol{q}}}(\boldsymbol{\lambda})\right)\Big)\Bigg] \tag{44}$$

$$= \sum_{\boldsymbol{q}} \left[B'(\boldsymbol{q}) \left(\log\left(f_{\boldsymbol{q}}(\boldsymbol{\lambda})\right) - \log\left(\sum_{\overline{\boldsymbol{q}}} f_{\overline{\boldsymbol{q}}}(\boldsymbol{\lambda})\right)\right)\right] + \sum_{\boldsymbol{q}} \left[B'(\boldsymbol{q}) \log\left(\sum_{\overline{\boldsymbol{q}}} f_{\overline{\boldsymbol{q}}}(\boldsymbol{\lambda})\right)\right] \tag{45}$$

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$$= \sum_{\boldsymbol{q}} \left[B'(\boldsymbol{q}) \log\left(B(\boldsymbol{q})\right)\right] + \log\left(\sum_{\overline{\boldsymbol{q}}} f_{\overline{\boldsymbol{q}}}(\boldsymbol{\lambda})\right) \sum_{\boldsymbol{q}} B'(\boldsymbol{q}) \quad (46)$$

$$= \sum_{\boldsymbol{q}} \left[B'(\boldsymbol{q}) \log\left(B(\boldsymbol{q})\right)\right] + \log\left(\sum_{\boldsymbol{q}} f_{\boldsymbol{q}}(\boldsymbol{\lambda})\right), \quad (47)$$

where

$$\sum_{\boldsymbol{q}} B'(\boldsymbol{q}) = \frac{\sum_{\boldsymbol{q}} f_{\boldsymbol{q}}(\boldsymbol{\lambda}')}{\sum_{\overline{\boldsymbol{q}}} f_{\overline{\boldsymbol{q}}}(\boldsymbol{\lambda}')} = 1 \quad (48)$$

and thus

$$Q(\boldsymbol{\lambda}', \boldsymbol{\lambda}') = \sum_{\boldsymbol{q}} \left[B'(\boldsymbol{q}) \log\left(B'(\boldsymbol{q})\right)\right] + \log\left(\sum_{\boldsymbol{q}} f_{\boldsymbol{q}}(\boldsymbol{\lambda}')\right). \quad (49)$$

By applying (47) and (49) and the definition of $\mathcal{F}(\boldsymbol{\lambda})$ in (8) to (42), we get the following expression

$$\sum_{\boldsymbol{q}} \left[B'(\boldsymbol{q}) \log\left(B(\boldsymbol{q})\right)\right] + \log\left(\sum_{\boldsymbol{q}} f_{\boldsymbol{q}}(\boldsymbol{\lambda})\right) - \sum_{\boldsymbol{q}} \left[B'(\boldsymbol{q}) \log\left(B'(\boldsymbol{q})\right)\right] - \log\left(\sum_{\boldsymbol{q}} f_{\boldsymbol{q}}(\boldsymbol{\lambda}')\right) - \log\left(\sum_{\boldsymbol{q}} f_{\boldsymbol{q}}(\boldsymbol{\lambda})\right) + \log\left(\sum_{\boldsymbol{q}} f_{\boldsymbol{q}}(\boldsymbol{\lambda}')\right) \overset{!}{\leq} 0. \quad (50)$$

We now need to prove the following inequality:

$$\sum_{\boldsymbol{q}} \left[B'(\boldsymbol{q}) \log\left(B(\boldsymbol{q})\right)\right] - \sum_{\boldsymbol{q}} \left[B'(\boldsymbol{q}) \log\left(B'(\boldsymbol{q})\right)\right] \overset{!}{\leq} 0 \quad (51)$$

or

$$\sum_{\boldsymbol{q}} B'(\boldsymbol{q}) \log\left(\frac{B(\boldsymbol{q})}{B'(\boldsymbol{q})}\right) \overset{!}{\leq} 0. \quad (52)$$

By using the fact that $(x-1) \geq \log(x)$, we can prove (52) by proving that

$$\sum_{\boldsymbol{q}} B'(\boldsymbol{q}) \left(\frac{B(\boldsymbol{q})}{B'(\boldsymbol{q})} - 1\right) \overset{!}{\leq} 0 \quad (53)$$

$$\sum_{\boldsymbol{q}} B(\boldsymbol{q}) - B'(\boldsymbol{q}) \overset{!}{\leq} 0 \quad (54)$$

$$\sum_{\boldsymbol{q}} B(\boldsymbol{q}) - \sum_{\boldsymbol{q}} B'(\boldsymbol{q}) \overset{!}{\leq} 0 \quad (55)$$

$$\sum_{\boldsymbol{q}} \frac{f_{\boldsymbol{q}}(\boldsymbol{\lambda})}{\sum_{\overline{\boldsymbol{q}}} f_{\overline{\boldsymbol{q}}}(\boldsymbol{\lambda})} - \sum_{\boldsymbol{q}} \frac{f_{\boldsymbol{q}}(\boldsymbol{\lambda}')}{\sum_{\overline{\boldsymbol{q}}} f_{\overline{\boldsymbol{q}}}(\boldsymbol{\lambda}')} \overset{!}{\leq} 0 \quad (56)$$

$$1 - 1 \overset{!}{\leq} 0. \quad (57)$$

This proves that $Q(\boldsymbol{\lambda}, \boldsymbol{\lambda}')$ is a strong sense auxiliary function for $\mathcal{F}(\boldsymbol{\lambda})$.

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